

OEM TECHNICAL MANUAL

CENTRIFUGAL LIQUID AMMONIA PUMPS

MODEL AP-900

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Facility: Vindhya Petrochemicals – Unit 3 (Ammonia Synthesis)

Target Tags: P-302A, P-302B

Equipment Type: Horizontal Single-Stage Centrifugal Liquid Ammonia Transfer Pumps

Document Classification: OEM Engineering Reference Manual

1. INTRODUCTION

This manual provides technical specifications, installation instructions, operating procedures, inspection requirements, maintenance practices, troubleshooting guidance, and overhaul criteria for AP-900 centrifugal liquid ammonia pumps installed at Vindhya Petrochemicals Unit 3.

The document applies specifically to:

Tag	Service
P-302A	Primary Liquid Ammonia Transfer Pump
P-302B	Standby Liquid Ammonia Transfer Pump

The AP-900 pump has been designed for continuous-duty transfer of condensed ammonia from downstream cooling and conditioning systems toward product storage service.

The equipment is intended for operation in demanding chemical process environments requiring high reliability, low vibration, and extended service intervals.

2. EQUIPMENT DESCRIPTION

2.1 General Design

The AP-900 is a horizontal centrifugal process pump employing:

- Single-stage radial-flow impeller
- Closed impeller design
- Centerline-mounted casing
- Mechanical seal arrangement
- Anti-corrosion wetted components
- Oil-lubricated rolling element bearings
- Flexible coupling drive system

The pump is designed for continuous operation under stable process conditions.

2.2 Functional Role

Within Unit 3, the AP-900 pumps receive liquid ammonia from process cooling sections and transfer product toward TK-401.

P-302A normally operates as the duty machine.

P-302B remains in standby service and shall be rotated periodically to ensure readiness.

2.3 Design Data

Parameter	Value
Pump Model	AP-900
Pump Type	Horizontal Centrifugal
Rated Flow	92 m ³ /hr
Rated Head	118 m
Rated Speed	2980 rpm
Design Pressure	32 bar

Hydrotest Pressure	48 bar
Design Temperature	-40°C to 85°C
Minimum Continuous Flow	27 m³/hr
Maximum Flow	120 m³/hr

3. TECHNICAL SPECIFICATIONS

3.1 Hydraulic Performance

The AP-900 hydraulic assembly is optimized near its best efficiency point (BEP).

Recommended operating range:

Parameter	Value
Minimum Flow	70% of BEP
Preferred Flow	90–110% of BEP
Maximum Flow	130% of BEP
BEP Efficiency	82%
NPSHr at BEP	3.6 m

Operation below minimum continuous flow may result in internal recirculation and accelerated wear.

3.2 Mechanical Specifications

Component	Material
Casing	ASTM A216 WCB
Impeller	ASTM A743 CF8M
Shaft	ASTM A479 410 SS
Wear Rings	Hardened SS
Seal Faces	Silicon Carbide

3.3 Driver Requirements

Parameter	Requirement
Voltage	415 VAC
Frequency	50 Hz
Rated Motor Power	75 kW
Full Load Current	128 A
Starting Method	Soft Starter
Power Factor	0.88

4. SAFETY INFORMATION

4.1 General Safety

Liquid ammonia presents risks associated with:

- Toxic exposure
- Frostbite injuries
- Pressure release events
- Mechanical hazards

Only trained personnel shall perform maintenance activities.

4.2 Isolation Requirements

Prior to maintenance:

1. Stop pump.
2. Isolate suction path.
3. Isolate discharge path.
4. Depressurize casing.
5. Drain residual liquid.
6. Lock out electrical power.

Failure to fully isolate the pump may result in serious injury.

4.3 Rotating Equipment Hazards

Do not remove:

- Coupling guards
- Bearing covers
- Inspection plates

while the pump remains in operation.

5. INSTALLATION REQUIREMENTS

5.1 Foundation Requirements

The foundation shall possess sufficient rigidity to prevent dynamic amplification.

Maximum permissible foundation flatness deviation:

0.10 mm per meter.

Minimum concrete curing period:

21 days.

5.2 Baseplate Alignment

Maximum allowable misalignment:

Parameter	Maximum
Angular Alignment	0.05 mm/100 mm
Parallel Offset	0.04 mm

Alignment shall be verified at operating temperature whenever practical.

5.3 Piping Requirements

The pump shall not support piping loads.

Maximum allowable nozzle loads:

Load Type	Limit
Axial Force	4.8 kN
Horizontal Force	3.5 kN
Vertical Force	3.5 kN
Resultant Moment	1.2 kN·m

6. PRE-START INSPECTION

6.1 Mechanical Checks

Verify:

- Coupling guard installed
 - Lubricant level correct
 - Fasteners secured
 - Rotation direction confirmed
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6.2 Fastener Torque Specifications

Component	Torque
Casing Bolts	115 Nm
Bearing Housing Bolts	62 Nm
Mechanical Seal Fasteners	18 Nm
Coupling Guard Bolts	12 Nm
Baseplate Anchor Nuts	175 Nm

Torque values shall be applied using calibrated tools.

6.3 Lubrication Requirements

Approved lubricant viscosity:

ISO VG 68

Bearing housing fill quantity:

2.4 liters

Oil level shall remain at centerline sight-glass mark.

7. STARTUP PROCEDURE

7.1 Initial Preparation

Verify:

- Pump casing filled.
- No trapped gas.
- Suction path established.
- Discharge path correctly aligned.

Dry running is prohibited.

7.2 Initial Start

Start motor and observe:

- Noise level
- Vibration
- Current draw
- Pressure response

Acceleration to rated speed should occur within:

8 seconds.

7.3 Stabilization Period

Monitor for:

15 minutes minimum.

Expected operating values:

Parameter	Typical Value
Speed	2980 rpm
Bearing Temperature	58–70°C
Motor Current	95–120 A
Vibration	1.5–3.0 mm/s RMS

8. NORMAL OPERATION

8.1 Operating Limits

Parameter	Limit
Maximum Bearing Temperature	85°C
Maximum Casing Temperature	90°C
Maximum Vibration	4.2 mm/s RMS
Maximum Motor Current	128 A
Maximum Differential Pressure	14 bar

8.2 Standby Pump Management

P-302B shall be operated periodically.

Recommended rotation schedule:

- Minimum 30 minutes
- Every 14 days

This prevents bearing brinelling and seal sticking.

9. CONDITION MONITORING

9.1 Vibration Monitoring

Measurement locations:

- Inboard bearing
- Outboard bearing
- Motor drive end
- Motor non-drive end

Alarm and trip limits:

Parameter	Alarm	Trip
Overall Vibration	4.2 mm/s RMS	6.0 mm/s RMS
Bearing Temperature	85°C	95°C

9.2 Mechanical Seal Monitoring

Acceptable seal leakage:

Less than 5 drops per minute.

Persistent leakage requires investigation.

9.3 Bearing Monitoring

Normal bearing temperature differential:

Less than 12°C between bearings.

Higher deviations may indicate internal damage.

10. PREVENTIVE MAINTENANCE

10.1 Monthly Inspection

Inspect:

- Bearing housing
 - Seal leakage
 - Coupling condition
 - Fastener tightness
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10.2 Quarterly Maintenance

Perform:

- Alignment verification
- Vibration survey
- Oil quality inspection

Acceptable oil contamination:

Less than ISO 18/16/13 cleanliness level.

10.3 Annual Maintenance

Replace:

- Bearing housing oil
- Mechanical seal elastomers
- Wear ring measurements

Review performance trends.

10.4 Major Overhaul

Recommended interval:

36,000 operating hours or 5 years.

Major overhaul includes:

- Complete disassembly
- Impeller inspection
- Shaft dimensional verification

- Bearing replacement
 - Seal replacement
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11. DISASSEMBLY PROCEDURE

11.1 Preparation

Confirm:

- Electrical isolation
 - Process isolation
 - Zero pressure condition
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11.2 Bearing Housing Removal

Remove fasteners gradually using cross-pattern sequence.

Uneven loading may distort housing surfaces.

11.3 Impeller Removal

Use approved puller assembly.

Direct hammer impact is prohibited.

Maximum permissible extraction force:

22 kN.

12. INSPECTION CRITERIA

12.1 Shaft Inspection

Replace shaft if:

- Runout exceeds 0.05 mm
 - Scoring exceeds 0.25 mm depth
 - Corrosion is visible in seal areas
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12.2 Impeller Inspection

Replace impeller if:

- Cracks are detected
 - Erosion exceeds 15% vane thickness
 - Dynamic balance cannot be restored
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12.3 Bearing Inspection

Reject bearings exhibiting:

- Discoloration
 - Brinelling
 - Pitting
 - Surface fatigue
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13. TROUBLESHOOTING

Symptom	Probable Cause	Corrective Action
Low Flow	Cavitation	Increase suction head
High Vibration	Misalignment	Realign equipment
High Bearing Temperature	Lubrication issue	Replace lubricant
Excess Current	Hydraulic overload	Reduce flow
Seal Leakage	Seal wear	Replace seal
Noise	Entrained vapor	Improve suction conditions

13.1 Cavitation Indicators

Common symptoms include:

- Crackling noise
- Flow instability
- Vibration increase
- Reduced hydraulic performance

Immediate corrective action is recommended.

14. ACCEPTANCE TESTING

Following maintenance, verify:

Test	Acceptance Requirement
Hydrostatic Test	No leakage
Rotation Test	Correct direction
Vibration Test	≤4.2 mm/s RMS
Bearing Temperature Test	≤85°C
Flow Verification	Within design range
Seal Leakage Test	Acceptable

Testing duration:

Minimum 60 minutes continuous operation.

15. SPARE PARTS STRATEGY

Minimum recommended inventory:

Component	Quantity
Mechanical Seal Kit	2
Bearing Set	2

Shaft Sleeve	1
Impeller	1
Wear Rings	2 Sets
Gasket Set	3 Sets

Critical spares shall be stored indoors between 10°C and 35°C.

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